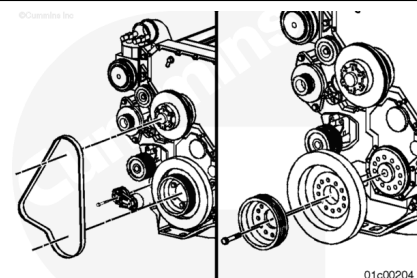


Trainings ▾

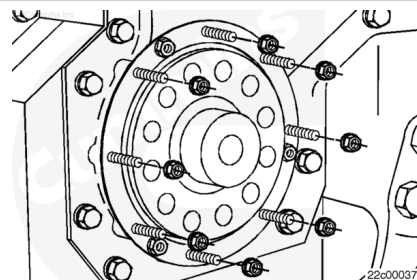
Preparatory Steps

- Refer to Procedure Remove the fan and water pump drive belt. 008-003 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-003.html)
- Refer to Procedure Remove the fan and water pump belt tensioner. 008-080 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-080.html)
- Refer to Procedure Remove the crankshaft pulley and vibration damper(s). 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-052-shopds.html)



Remove

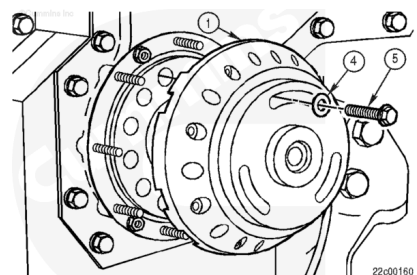
Remove the eight mounting nuts from the gear cover studs.



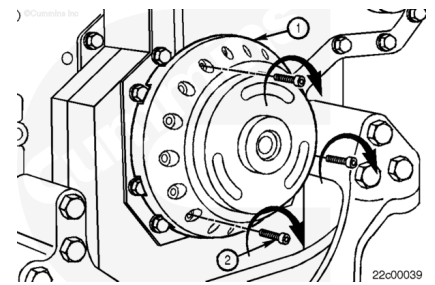
Position the replacer plate (1) over the crankshaft end and against the crankshaft seal. The large notch between the teeth needs to bridge the raised gear cover surface.

Make certain that the alignment of the three welded nuts on the crankshaft seal are positioned into the counterbore slots of the replacer plate (1).

Install a capscrew (4) with a flat washer (5) in the upper slot of the replacer plate to hold it in position.



Tighten the capscrews by hand until a slight resistance is felt. Do not overtighten the capscrews or the seal nuts can break off causing damage to the seal and leading to front cover damage.



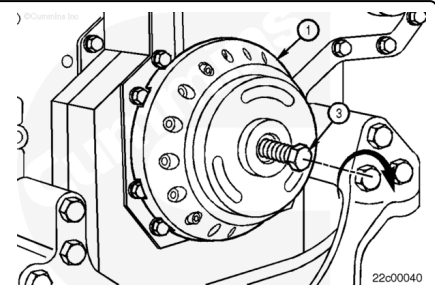
Use a small allen wrench to thread the removal tool capscrews (2) through the replacer plate and into the seal nuts.

Tighten the socket head capscrews (2) **clockwise** into the three seal removal nuts that are welded on the crankshaft seal until a slight resistance is felt.

Remove the capscrew in the upper slot of the replacer plate (1).

Install the replacer screw (3) into the replacer plate (1).

Turn the screw (3) **clockwise** until the crankshaft seal is removed from the crankshaft.



Clean and Inspect for Reuse

Note : All lower gear covers have either a beveled half groove, or a full o-ring groove under the crankshaft seal flange.

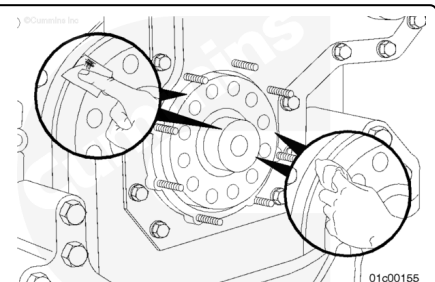
Full o-ring groove covers contain an o-ring in the lower gear cover groove and were **only** used prior to June 5, 1999. The lower gear cover **must** be replaced if installing a new seal. Refer to Procedure 001-080 in Section 1.

(/qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html)

Beveled half groove covers do **not** contain an o-ring and **only** require replacement if damaged. Refer to Procedure 001-080 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-080-tr.html)

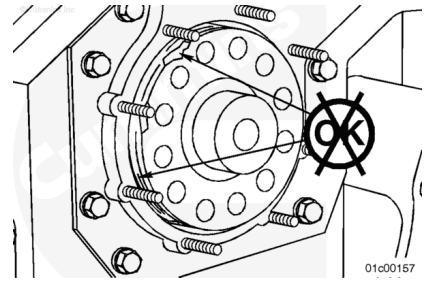
Use a small magnet to remove any metal shavings.

Use a crocus cloth to remove any deposits from the crankshaft adaptor and seal bore area.



Inspect the crankshaft adaptor for nicks, burrs, and grooves. Damage to the sealing surface can result in seal leakage.

If a fingernail catches in a nick, burr, or groove, the crankshaft or crankshaft adaptor **must** be replaced.



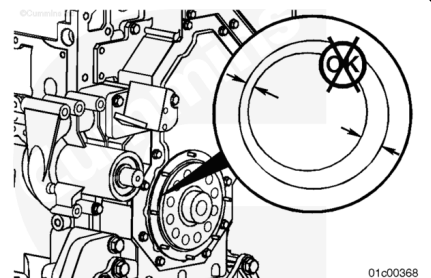
Install

The position of the front crankshaft seal on the crankshaft is determined by the position of the lower gear cover.

The lower gear cover is mounted to the gear cover housing using clearance holes. This allows for slight movement of the cover when the mounting capscrews are loose.

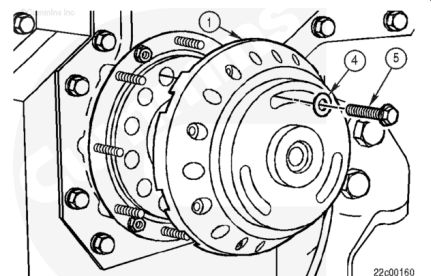
The unitized crankshaft seal **must** be centered on the crankshaft for correct operation.

The seal attaches to the lower gear cover, and **must** be installed with the lower gear cover capscrews loose and the cover free to move.



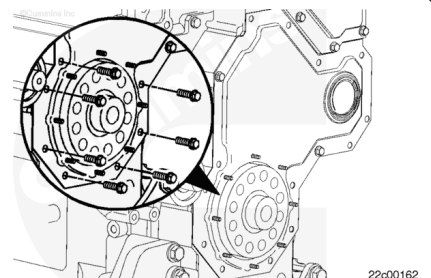
In order for the seal to be properly centered on the crankshaft during assembly and maintain its location during operation, the following installation instructions **must** be followed.

The crankshaft seal removal and installation tool correctly centers the seal and correctly positions the dust seal portion of the seal on the crankshaft.



⚠ CAUTION ⚠

Failure to loosen and remove the mounting capscrews from the gear cover can cause the front crankshaft seal to be misaligned and cause premature seal wear.



The front gear cover **must** be aligned as follows:

Removing the gear cover mounting capscrews around the crankshaft seal allows the replacer plate to set against the gear cover during seal installation.

Remove the gear cover mounting capscrews around the crankshaft seal.

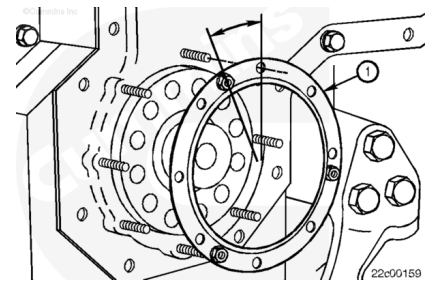
Loosen the remaining mounting capscrews on the gear cover so the cover can move freely.

⚠ CAUTION ⚠

Failure to follow the installation and tightening sequence can lead to the crankshaft seal being off center and installed at the wrong depth causing premature seal wear.

⚠ CAUTION ⚠

Never use a hammer or mallet to install the seal. Use of a hammer can damage the crankshaft seal and/or crankshaft.



The seal removal and installation tool has provisions for any seal nut orientation, although one seal nut **must** always be located at the 11 o'clock position.

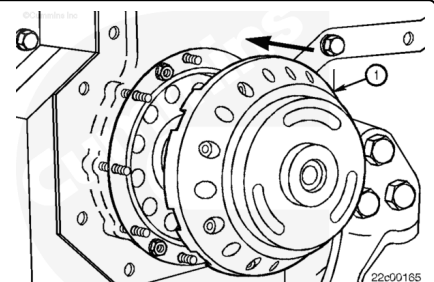
Do **not** use lubricants on the crankshaft seal.

Position the crankshaft seal (1) onto the crankshaft until the gear cover studs are protruding through the seal carrier mounting holes.

Place the replacer plate (1) against the crankshaft seal.

Make sure the welded nuts on the front crankshaft seal are seated in their respective slots on the replacer plate (1).

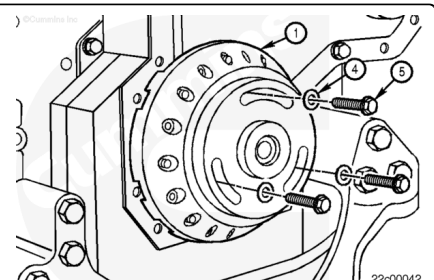
The large notch between the teeth **must** bridge the raised gear cover surface.



Install three washers (4) and capscrews (5) through the three slots in the replacer plate (1) and into the crankshaft end.

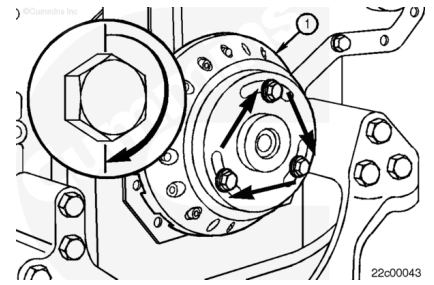
The capscrews in the replacer plate **must** be equally spaced. This allows the crankshaft seal to be installed correctly.

Tighten the capscrews hand-tight.



⚠ CAUTION ⚠

The crankshaft seal must be pressed evenly. If the crankshaft seal is not pressed in evenly the seal can be damaged.



Inspect the crankshaft seal during installation to make certain it is being pressed evenly.

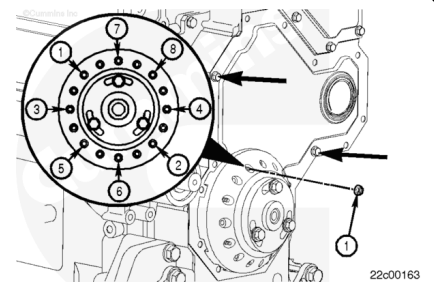
Tighten the capscrew(s) no more than $\frac{1}{2}$ of a revolution at a time, alternating in a **clockwise** pattern until the replacer plate (1) bottoms out against the face of the lower gear cover.

Torque Value: 20 n•m [177 in-lb]

To maintain the cover position while the replacer plate is removed, tighten the two gear cover mounting capscrews, as indicated by the arrows in the graphic.

Install the crankshaft seal mounting nuts (1) through the holes provided in the replacer plate.

Tighten the mounting nuts in the sequence shown.



Torque Value: 8 n•m [71 in-lb]

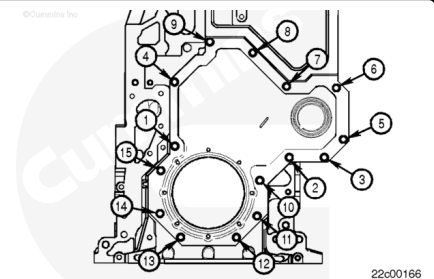
Remove the replacer plate.

Install all the gear cover mounting capscrews.

The torque sequence **must** be followed to prevent the gear cover strip seal from squeezing out of the groove.

Tighten the capscrews in the sequence shown.

Torque Value: 28 n•m [248 in-lb]



Finishing Steps

- Refer to Procedure Install the crankshaft pulley and vibration damper(s). 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html>)
- Refer to Procedure Install the fan and water pump belt tensioner. 008-080 in Section 8.

(/qs3/pubsys2/xml/en/procedures/10/10-008-080-tr.html)

- Refer to Procedure Install the fan and water pump drive belt. 008-003 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-003.html)

